

# EMBODIED ABSTRACTION FAILURE MODES: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

**Anonymous authors**

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## ABSTRACT

We study abstraction in robot planning. The motivating bet is: Find where symbolic or language abstractions erase action-critical mechanics. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

## 1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For abstraction in robot planning, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

## 2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2025a; 2023; 2024; 2026a;b; 2025b). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

## 3 MECHANISM

Given state  $s$  and action  $a$ , the model returns an observed next-state prediction plus a finite atlas  $B(s, a)$  of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

## 4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

| Variant                   | Split             | Success | Tail risk |
|---------------------------|-------------------|---------|-----------|
| observed_only_wam         | nominal           | 0.520   | 0.480     |
| observed_only_wam         | hidden_mode_shift | 0.198   | 0.802     |
| augmentation_as_data_wam  | nominal           | 0.610   | 0.390     |
| augmentation_as_data_wam  | hidden_mode_shift | 0.245   | 0.755     |
| scalar_uncertainty_wam    | nominal           | 0.652   | 0.348     |
| scalar_uncertainty_wam    | hidden_mode_shift | 0.323   | 0.677     |
| proposed_branch_mechanism | nominal           | 0.755   | 0.245     |
| proposed_branch_mechanism | hidden_mode_shift | 0.545   | 0.455     |

## 5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

## 6 CONCLUSION

Embodied Abstraction Failure Modes argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

## REFERENCES

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